The documentation and process conversion measures necessary to comply with this revision shall be completed by 11 April 2009.

INCH - POUND

MIL-PRF-19500/570D
11 January 2009
SUPERSEDING
MIL-PRF-19500/570C
17 August 2005

PERFORMANCE SPECIFICATION SHEET

SEMICONDUCTOR DEVICE, TRANSISTOR, FIELD EFFECT,
N-CHANNEL, SILICON LOGIC-LEVEL, TYPES 2N6901 AND 2N6903,
JAN, JANTX, JANTXV, AND JANS

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and MIL-PRF-19500.

1. SCOPE

1.1 Scope. This specification covers the performance requirements for a logic-level N-channel, enhancement-mode, MOSFET, power transistor. Four levels of product assurance are provided for each device type as specified in MIL-PRF-19500.

1.2 Physical dimensions. See figure 1, TO-205AF (formerly TO-39).

1.3 Maximum ratings. Unless otherwise specified, $T_A$ = +25°C.

| Type | $P_T$ (1) $T_C$ = +25°C | $P_T$ $T_A$ = +25°C | $R_{\theta JC}$ (2) | $V_{DS}$ | $V_{DG}$ | $V_{GS}$ | $I_{D1}$ (3) $T_C$ = +25°C | $I_{D2}$ (3) $T_C$ = +100°C | $I_S$ | $I_{DM}$ | $T_J$ and $T_{STG}$ |
|---|---|---|---|---|---|---|---|---|---|---|---|
| | W | W | °C /W | V dc | V dc | V dc | A dc | A dc | A dc | A(pk) | °C |
| 2N6901 | 8.33 | 0.6 | 15.0 | 100 | 100 | ±10 | 1.69 | 1.07 | 1.69 | 5 | -55 to +150 |
| 2N6903 | 8.33 | 0.6 | 15.0 | 200 | 200 | ±10 | 0.98 | 0.62 | 0.98 | 4 | |

(1) Derated linearly by 0.067 W/°C for $T_C$ > +25°C.
(2) See figure 2, thermal impedance curves.
(3) The following formula derives the maximum theoretical $I_D$ limit. $I_D$ is limited by internal construction and may be limited by pin diameter:

$$I_D = \sqrt{\frac{T_{JM} - T_C}{(R_{\theta JC}) \times (R_{DS}(on) \text{ at } T_{JM})}}$$

Comments, suggestions, or questions on this document should be addressed to Defense Supply Center, Columbus, ATTN: DSCC-VAC, P.O. Box 3990, Columbus, OH 43218-3990, or emailed to Semiconductor@dscc.dla.mil. Since contact information can change, you may want to verify the currency of this address information using the ASSIST Online database at http://assist.daps.dla.mil .

AMSC N/A

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1.4 Primary electrical characteristics. Unless otherwise specified, at $T_C$ = +25°C.

| Type | Min $V_{(BR)DSS}$ $V_{GS}$ = 0 V $I_D$ = 1 mA | $V_{GS(th)1}$ $V_{DS} \geq V_{GS}$ $I_D$ = 1 mA | Max $I_{DSS1}$ $V_{GS}$ = 0 $V_{DS}$ = 80 percent of rated $V_{DS}$ | Max $r_{DS(on)}$ (1) $V_{GS}$ = 5 V dc | |
|---|---|---|---|---|---|
| | | | | $T_J$ = +25°C at $I_{D1}$ | $T_J$ = +150°C at $I_D2$ |
| | V dc | V dc | µA dc | Ohms | Ohms |
| | | Min Max | | | |
| 2N6901 | 100 | 1.0 2.0 | 1.0 | 1.4 | 2.9 |
| 2N6903 | 200 | 1.0 2.0 | | 3.65 | 8.65 |

(1) Pulsed (see 4.5.1).

2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3, 4, or 5 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements of documents cited in sections 3, 4, or 5 of this specification, whether or not they are listed.

2.2 Government documents.

2.2.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

DEPARTMENT OF DEFENSE SPECIFICATIONS

MIL-PRF-19500 - Semiconductor Devices, General Specification for.

DEPARTMENT OF DEFENSE STANDARDS

MIL-STD-750 - Test Methods for Semiconductor Devices.

(Copies of these documents are available online at http://assist.daps.dla.mil/quicksearch/ or http://assist.daps.dla.mil or from the Standardization Document Order Desk, 700 Robbins Avenue, Building 4D, Philadelphia, PA 19111-5094.)

* 2.3 Order of precedence. Unless otherwise noted herein or in the contract, in the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.



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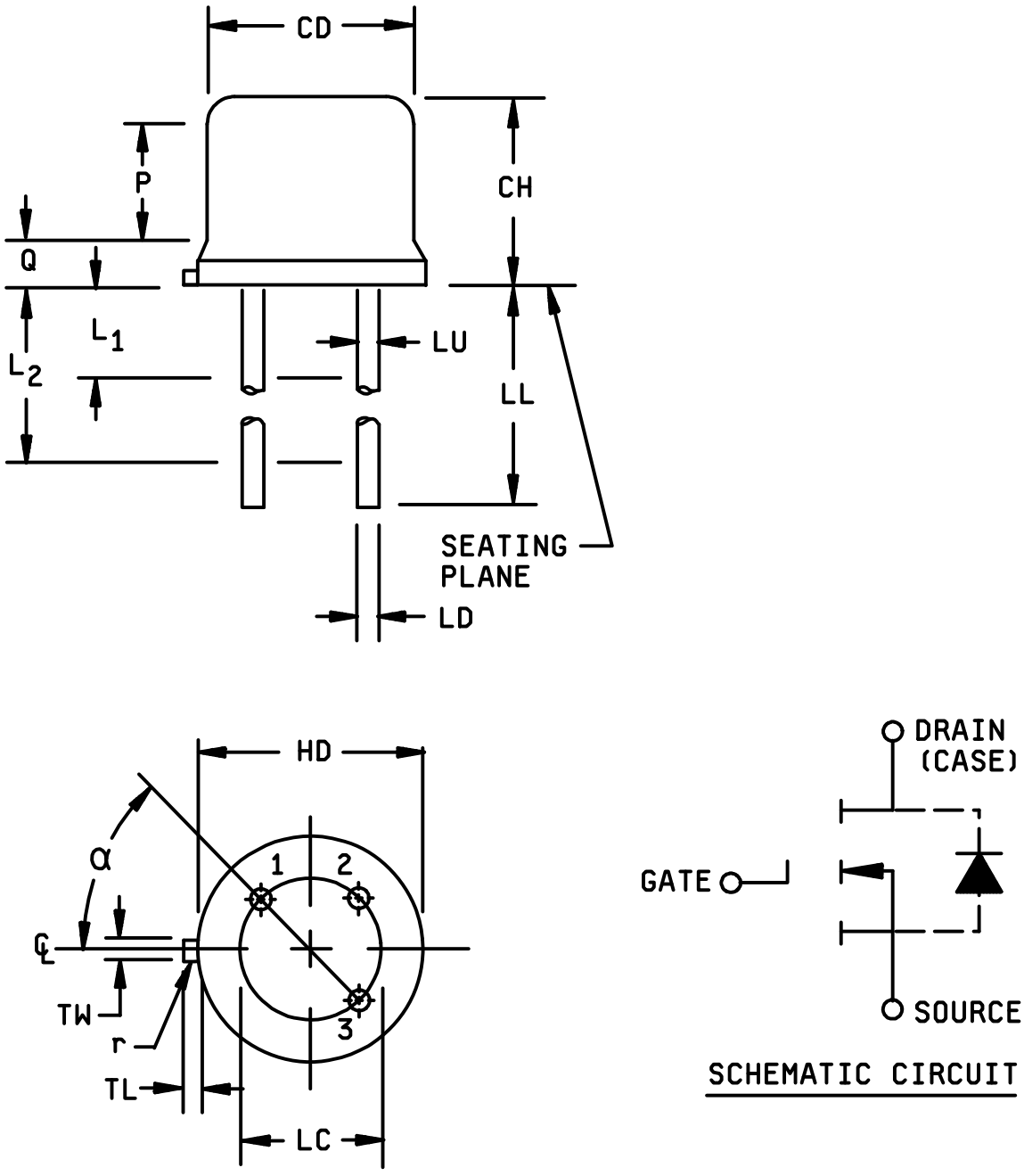



FIGURE 1. Physical dimensions for TO-205 AF.



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| Dimensions | | | | | |
|---|---|---|---|---|---|
| Ltr | Inches | | Millimeters | | Notes |
| | Min | Max | Min | Max | |
| CD | .305 | .335 | 7.75 | 8.51 | |
| CH | .160 | .180 | 4.07 | 4.57 | |
| HD | .335 | .370 | 8.51 | 9.40 | |
| LC | .200 TP | | 5.08 TP | | |
| LD | .016 | .021 | 0.41 | 0.53 | 8,9 |
| LL | .500 | .750 | 12.70 | 19.05 | 8,9 |
| LU | .016 | .019 | 0.41 | 0.48 | 8,9 |
| $L_1$ | | .050 | | 1.27 | 8,9 |
| $L_2$ | .250 | | 6.35 | | 8,9 |
| P | .100 | | 2.54 | | 6 |
| Q | | .050 | | 1.27 | 5 |
| TL | .029 | .045 | 0.74 | 1.14 | 4 |
| TW | .028 | .034 | 0.71 | 0.86 | 3 |
| r | | .010 | | 0.25 | 10 |
| $\alpha$ | 45° TP | | 45° TP | | 6 |

NOTES:
1. Dimensions are in inches.
2. Millimeters are given for general information only.
3. Beyond radius(r) maximum, TW shall be held for a minimum length of .011 (0.28 mm).
4. Dimension TL measured from maximum HD.
5. Outline in this zone is not controlled.
6. Dimension CD shall not vary more than .010 (0.25 mm) in zone P. This zone is controlled for automatic handling.
7. Leads at gauge plane .054 +.001, -.000 (1.37 +0.03, -0.00 mm) below seating plane shall be within .007 (0.18 mm) radius of true position (TP) at maximum material condition (MMC) relative to tab at MMC.
8. LU applies between $L_1$ and $L_2$. LD applies between $L_2$ and LL minimum. Diameter is uncontrolled in $L_1$ and beyond LL minimum.
9. All three leads.
10. Radius(r) applies to both inside corners of tab.
11. Drain is electrically connected to the case.
12. Pin out: 1- source, 2 - gate, 3 - drain (case).
13. In accordance with ASME Y14.5M, diameters are equivalent to $\phi$x symbology.

FIGURE 1. Physical dimensions for TO-205 AF - Continued.



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3. REQUIREMENTS

3.1 General. The individual item requirements shall be as specified in MIL-PRF-19500 and as modified herein.

3.2 Qualification. Devices furnished under this specification shall be products that are manufactured by a manufacturer authorized by the qualifying activity for listing on the applicable qualified manufacturers list before contract award (see 4.2 and 6.3).

3.3 Abbreviations, symbols, and definitions. Abbreviations, symbols, and definitions used herein shall be as specified in MIL-PRF-19500 and as follows:

nC .....nano Coulomb

3.4 Interface and physical dimensions. Interface and physical dimensions shall be as specified in MIL-PRF-19500, and on figure 1.

3.4.1 Lead finish. Lead finish shall be solderable in accordance with MIL-PRF-19500, MIL-STD-750, and herein. Where a choice of lead finish is desired, it shall be specified in the acquisition document (see 6.2).

3.5 Marking. Marking shall be in accordance with MIL-PRF-19500.

3.6 Electrical performance characteristics. Unless otherwise specified herein, the electrical performance characteristics are as specified in 1.3, 1.4, and table I herein.

3.7 Electrical test requirements. The electrical test requirements shall be as specified in table I.

3.8 Electrostatic discharge (ESD) protection. The devices covered by this specification require electrostatic discharge protection (see 3.8.1).

3.8.1 Handling. Metal oxide semiconductor (MOS) devices must be handled with certain precautions to avoid damage due to the accumulation of static charge. However, the following handling practices are recommended (see 3.8).

a. Devices should be handled on benches with conductive handling devices.

b. Ground test equipment, tools, and personnel handling devices.

c. Do not handle devices by the leads.

d. Store devices in conductive foam or carriers.

e. Avoid use of plastic, rubber or silk in MOS areas.

f. Maintain relative humidity above 50 percent if practical.

g. Care should be exercised during test and troubleshooting to apply not more than maximum rated voltage to any lead.

h. Gate must be terminated to source, R $\leq$ or 100 k$\Omega$, whenever bias voltage is applied drain to source.

3.9 Workmanship. Semiconductor devices shall be processed in such a manner as to be uniform in quality and shall be free from other defects that will affect life, serviceability, or appearance.



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4. VERIFICATION

4.1 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. Qualification inspection (see 4.2).

b. Screening (see 4.3).

c. Conformance inspection (see 4.4 and tables I and II).

4.2 Qualification inspection. Qualification inspection shall be in accordance with MIL-PRF-19500 and as specified herein.

4.2.1 Group E qualification. Group E inspection shall be performed for qualification or re-qualification only. In case qualification was awarded to a prior revision of the specification sheet that did not request the performance of table II tests, the tests specified in table II herein that were not performed in the prior revision shall be performed on the first inspection lot of this revision to maintain qualification.



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* 4.3 Screening (JANS, JANTX, and JANTXV levels only). Screening shall be in accordance with table E-IV of MIL-PRF-19500, and as specified herein. The following measurements shall be made in accordance with table I herein. Devices that exceed the limits of table I herein shall not be acceptable.

| Screen (see, table E-IV of MIL-PRF-19500) (1) (2) | Measurement | |
|---|---|---|
| | JANS level | JANTX and JANTXV levels |
| (3) | Gate stress test (see 4.3.1) | Gate stress test (see 4.3.1) |
| (3) | Method 3470 of MIL-STD-750, (see 4.3.2) optional | Method 3470 of MIL-STD-750, (see 4.3.2) optional |
| (3) 3c | Method 3161 of MIL-STD-750, (see 4.3.3) | Method 3161 of MIL-STD-750, (see 4.3.3) |
| 9 | $I_{GSSF1}$, $I_{GSSR1}$, $I_{DSS1}$, subgroup 2 of table I herein | Not applicable |
| 10 | Method 1042 of MIL-STD-750, test condition B | Method 1042 of MIL-STD-750, test condition B |
| 11 | Subgroup 2 of table I herein;<br>$I_{GSSF1}$, $I_{GSSR1}$, $I_{DSS1}$, $r_{DS(on)1}$, $V_{GS(th)1}$<br>$\Delta I_{GSSF1}$ = ±20 nA dc or ±100 percent of initial value, whichever is greater.<br>$\Delta I_{GSSR1}$ = ±20 nA dc or ±100 percent of initial value, whichever is greater.<br>$\Delta I_{DSS1}$ = ±0.2 μA dc or ±100 percent of initial value, whichever is greater. | Subgroup 2 of table I herein<br>$I_{GSSF1}$, $I_{GSSR1}$, $I_{DSS1}$, $r_{DS(on)1}$, $V_{GS(th)1}$ |
| 12 | Method 1042 of MIL-STD-750, test condition A, t = 240 hours | Method 1042 of MIL-STD-750, test condition A or t = 48 hours minimum at+175°C minimum. |
| 13 | Subgroups 2 and 3 of table I herein;<br>$\Delta I_{GSSF1}$ = ±20 nA dc or ±100 percent of initial value, whichever is greater.<br>$\Delta I_{GSSR1}$ = ±20 nA dc or ±100 percent of initial value, whichever is greater.<br>$\Delta I_{DSS1}$ = ±0.2 μA dc or ±100 percent of initial value, whichever is greater<br>$\Delta r_{DS(on)1}$ = ±20 percent of initial value.<br>$\Delta V_{GS(th)1}$ = ±20 percent of initial value. | Subgroup 2 of table I herein;<br>$\Delta I_{GSSF1}$ = ±20 nA dc or ±100 percent of initial value, whichever is greater.<br>$\Delta I_{GSSR1}$ = ±20 nA dc or ±100 percent of initial value, whichever is greater.<br>$\Delta I_{DSS1}$ = ±0.2 μA dc or ±100 percent of initial value, whichever is greater.<br>$\Delta r_{DS(on)1}$ = ±20 percent of initial value.<br>$\Delta V_{GS(th)1}$ = ±20 percent of initial value. |

(1) At the end of the test program, $I_{GSSF1}$, $I_{GSSR1}$, and $I_{DSS1}$ are measured.
(2) An out-of-family program to characterize $I_{GSSF1}$, $I_{GSSR1}$, $I_{DSS1}$, and $V_{GS(th)1}$ shall be invoked.
(3) Shall be performed anytime after temperature cycling, screen 3a; and does not need to be repeated in screening requirements.



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4.3.1 Gate stress test. Apply $V_{GS}$ = 15 V min. for t = 250 μs min.

4.3.2 Unclamped inductive switching.

a. Peak current, $I_D$ .....rated $I_{D1}$.

b. Peak gate voltage, $V_{GS}$ .....10 V.

c. Gate to source resistor, $R_{GS}$ .....25 W < $R_{GS}$ < 200 W.

d. Initial case temperature .....+25°C, +10°C -5°C.

e. Inductance .....100 μH ±10 percent.

f. Number of pulses to be applied .....1 pulse.

* 4.3.3 Thermal impedance. The thermal impedance measurements shall be performed in accordance with method 3161 of MIL-STD-750 using the guidelines in that method for determining $I_M$, $I_H$, $t_H$, $t_{SW}$, ($V_C$ and $V_H$ where appropriate). Measurement delay time ($t_{MD}$) = 70 μs max. See table II, group E, subgroup 4 herein.

4.4 Conformance inspection. Conformance inspection shall be in accordance with MIL-PRF-19500 and as specified herein.

4.4.1 Group A inspection. Group A inspection shall be conducted in accordance with table E-V of MIL-PRF-19500, and table I herein. Electrical measurements (end-points) shall be in accordance with table I, subgroup 2 herein.

4.4.2 Group B inspection. Group B inspection shall be conducted in accordance with the conditions specified for subgroup testing in appendix E, table E-VIA (JANS) and table E-VIB (JAN, JANTX, and JANTXV) of MIL-PRF-19500, and as follows. Electrical measurements (end-points) shall be in accordance with table I, subgroup 2 herein.

4.4.2.1 Group B inspection, appendix E, table E-VIA (JANS) of MIL-PRF-19500.

| | Subgroup | Method | Conditions |
|---|---|---|---|
| * | B3 | 1051 | Test condition G. |
| | B3 | 2077 | SEM. |
| | B4 | 1042 | Test condition D: 2,000 cycles. The heating cycle shall be 1 minute minimum. |
| | B5 | 1042 | Test condition A, $V_{DS}$ = rated, $T_A$ = +175°C, t = 120 hours. |
| | B5 | 1042 | Test condition B, $V_{GS}$ = rated, $T_A$ = +175°C, t = 24 hours. |
| * | B5 | 2037 | Test condition D. |

4.4.2.2 Group B inspection, appendix E, table E-VIB (JAN, JANTX, and JANTXV) of MIL-PRF-19500.

| Subgroup | Method | Conditions |
|---|---|---|
| B2 | 1051 | Test condition G. |
| B3 | 1042 | Test condition D: 2,000 cycles. The heating cycle shall be 30 seconds minimum. |



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4.4.3 Group C inspection. Group C inspection shall be conducted in accordance with the conditions specified for subgroup testing in appendix E, table E-VII of MIL-PRF-19500, and as follows. Electrical measurements (end-points) shall be in accordance with table I, subgroup 2 herein.

| | Subgroup | Method | Conditions |
|---|---|---|---|
| | C2 | 2036 | Test condition E. |
| * | C5 | 3161 | See 4.5.2, $R_{\theta JC}$(max) = 15°C /W. |
| | C6 | 1042 | Test condition D: 6,000 cycles. The heating cycle shall be 30 seconds minimum. |

4.4.4 Group E inspection. Group E inspection shall be conducted in accordance with the conditions specified for subgroup testing in appendix E, table E-IX of MIL-PRF-19500, and table II herein. Electrical measurements (end-points) shall be in accordance with table I, subgroup 2 herein.

4.5 Methods of inspection. Methods of inspection shall be as specified in the appropriate tables and as follows:

4.5.1 Pulse measurements. Conditions for pulse measurements shall be as specified in section 4 of MIL-STD-750.

* 4.5.2 Thermal resistance. The thermal resistance measurements shall be performed in accordance with method 3161 of MIL-STD-750 using the guidelines in that method for determining $I_M$, $I_H$, $t_H$, $t_{SW}$ ($V_C$ and $V_H$ where appropriate). Measurement delay time ($t_{MD}$) = 70 μs max. See table E-IX of MIL-PRF-19500, group E, subgroup 4.



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* TABLE I. Group A inspection.

| Inspection 1/ | MIL-STD-750 | | Symbol | Limits | | Unit |
|---|---|---|---|---|---|---|
| | Method | Conditions | | Min | Max | |
| Subgroup 1 | | | | | | |
| Visual and mechanical inspection | 2071 | | | | | |
| Subgroup 2 | | | | | | |
| * Thermal impedance 2/ | 3161 | See 4.3.3 | $Z_{\theta JC}$ | | | °C /W |
| Breakdown voltage drain to source | 3407 | $I_D$ = 1.0 mA dc, bias condition C, $V_{GS}$ = 0 | $V_{(BR)DSS}$ | | | |
| 2N6901<br>2N6903 | | | | 100<br>200 | | V dc<br>V dc |
| Gate to source voltage (threshold) | 3403 | $V_{DS} \geq V_{GS}$, $I_D$ = 1.0 mA dc | $V_{GS(th)1}$ | 1.0 | 2.0 | V dc |
| Gate current | 3411 | $V_{GS}$ = +10 V dc, bias condition C, $V_{DS}$ = 0 | $I_{GSSF1}$ | | +100 | nA dc |
| Gate current | 3411 | $V_{GS}$ = -10 V dc, bias condition C, $V_{DS}$ = 0 | $I_{GSSR1}$ | | -100 | nA dc |
| Drain current<br>2N6901<br>2N6903 | 3413 | Bias condition C, $V_{GS}$ = 0<br>$V_{DS}$ = 80 V dc<br>$V_{DS}$ = 160 V dc | $I_{DSS1}$ | | <br>1.0<br>1.0 | <br>µA dc<br>µA dc |
| Static drain to source on-state resistance<br>2N6901<br>2N6903 | 3421 | $V_{GS}$ = 5 V dc, bias condition A, pulsed (see 4.5.1)<br>$I_D$ = 1.07 A dc<br>$I_D$ = 0.62 A dc | $r_{DS(on)1}$ | | <br>1.4<br>3.65 | <br>Ω<br>Ω |
| Drain to source on-state voltage<br>2N6901<br>2N6903 | 3405 | $V_{GS}$ = 5 V dc, bias condition A, pulsed (see 4.5.1)<br>$I_D$ = 1.69 A dc<br>$I_D$ = 0.98 A dc | $V_{DS(on)}$ | | <br><br>2.4<br>6.0 | <br><br>V<br>V |
| Forward voltage (source drain diode)<br>2N6901<br>2N6903 | 4011 | Pulsed (see 4.5.1), $V_{GS}$ = 0<br><br>$I_S$ = 1.69 A dc<br>$I_S$ = .98 A dc | $V_{SD}$ | 0.8 | 1.6 | V |
| Forward transconductance | 3475 | $I_D$ = rated $I_{D2}$ (see 1.3), pulsed (see 4.5.1) | $g_{FS}$ | 0.5 | 2.0 | S |

See footnote at end of table.



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* TABLE I. Group A inspection - Continued.

| Inspection 1/ | MIL-STD-750 | | Symbol | Limits | | Unit |
|---|---|---|---|---|---|---|
| | Method | Conditions | | Min | Max | |
| Subgroup 3 | | | | | | |
| High temperature operation: | | $T_C = T_J = +125°C$ | | | | |
| Gate to source voltage (threshold) | 3403 | $V_{DS} \geq V_{GS}$, $I_D$ = 1.0 mA dc | $V_{GS(th)2}$ | 0.5 | | V dc |
| Gate current | 3411 | $V_{GS}$ = +10 V dc and -10 V dc; $V_{DS}$ = 0; bias condition C | $I_{GSS2}$ | | ±200 | nA dc |
| Drain current | 3413 | Bias condition C, $V_{GS}$ = 0 V | $I_{DSS2}$ | | 50 | µA dc |
| 2N6901<br>2N6903 | | $V_{DS}$ = 80 V dc<br>$V_{DS}$ = 160 V dc | | | | |
| Static drain to source on-state resistance | 3421 | $V_{GS}$ = 5 V dc, pulsed (see 4.5.1) | $r_{DS(on)2}$ | | | |
| 2N6901<br>2N6903 | | $I_D$ = 1.07 A dc<br>$I_D$ = 0.62 A dc | | | 2.6<br>7.7 | Ω<br>Ω |
| Low temperature operation: | | $T_C = T_J = -55°C$ | | | | |
| Gate to source voltage (threshold) | 3403 | $V_{DS} \geq V_{GS}$; $I_D$ = 1.0 mA dc | $V_{GS(th)3}$ | | 3.0 | V dc |
| Subgroup 4 | | | | | | |
| Switching time test | 3472 | $I_D$ = rated $I_{D2}$, (see 1.3); $V_{GS}$ = 5 V dc, gate drive impedance = 25 Ω | | | | |
| Turn-on delay time | | | $t_{d(on)}$ | | 25 | ns |
| 2N6901<br>2N6903 | | $V_{DD}$ = 50 V dc<br>$V_{DD}$ = 100 V dc | | | | |
| Rise time | | | $t_r$ | | 80 | ns |
| 2N6901<br>2N6903 | | $V_{DD}$ = 50 V dc<br>$V_{DD}$ = 100 V dc | | | | |

See footnote at end of table.



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* TABLE I. Group A inspection - Continued.

| | Inspection 1/ | MIL-STD-750 | | Symbol | Limits | | Unit |
|---|---|---|---|---|---|---|---|
| | | Method | Conditions | | Min | Max | |
| | Subgroup 4 - Continued | | | | | | |
| | Turn-off delay time | | | $t_{d(off)}$ | | | |
| | 2N6901 | | $V_{DD}$ = 50 V dc | | | 45 | ns |
| | 2N6903 | | $V_{DD}$ = 100 V dc | | | 40 | ns |
| | Fall time | | | $t_f$ | | 80 | ns |
| | 2N6901 | | $V_{DD}$ = 50 V dc | | | | |
| | 2N6903 | | $V_{DD}$ = 100 V dc | | | | |
| | Subgroup 5 | | | | | | |
| | Safe operating area (SOA) test | 3474 | See figure 3<br>$V_{DS}$ = 80 percent of rated $V_{DS}$ and $V_{DS} \leq$ 200 V max | | | | |
| | High voltage dc SOA | | $t_p$ = 1 ms | | | | |
| | Electrical measurements | | See table I, subgroup 2 | | | | |
| | Single pulse unclamped inductive switching | 3470 | See 4.3.2; c = 0, 116 devices | | | | |
| | Electrical measurements | | See table I, subgroup 2 | | | | |
| | Subgroups 6 | | | | | | |
| | Not applicable | | | | | | |
| | Subgroup 7 | | | | | | |
| | Gate charge | 3471 | Condition A or B | | | | |
| | On-state gate charge | | | $Q_{g(on)}$ | | | |
| * | 2N6901 | | | | | 5 | nC |
| * | 2N6903 | | | | | 5 | nC |

See footnote at end of table.



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* TABLE I. Group A inspection - Continued.

| | Inspection 1/ | MIL-STD-750 | | Symbol | Limits | | Unit |
|---|---|---|---|---|---|---|---|
| | | Method | Conditions | | Min | Max | |
| | Subgroup 7 - Continued | | | | | | |
| | Gate to source charge | | | $Q_{gs}$ | | | |
| * | 2N6901 | | | | | 1.0 | nC |
| * | 2N6903 | | | | | 0.8 | nC |
| | Gate to drain charge | | | $Q_{gd}$ | | | |
| * | 2N6901 | | | | | 2.9 | nC |
| * | 2N6903 | | | | | 2.7 | nC |
| | Reverse recovery time | 3473 | $V_{DD} \leq 30$ V; di/dt = 100 A/μs<br>$I_F$ = 1 A | $t_{rr}$ | | | |
| | 2N6901 | | | | | 250 | ns |
| | 2N6903 | | | | | 500 | ns |

1/ For sampling plan, see MIL-PRF-19500.
2/ This test required for the following end-point measurements only:
Group B, subgroups 3 and 4 (JANS).
Group B, subgroups 2 and 3 (JAN, JANTX, and JANTXV).
Group C, subgroup 2 and 6.
Group E, subgroup 1.



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* TABLE II. Group E inspection (all quality levels) for qualification or re-qualification only.

| Inspection | MIL-STD-750 | | Qualification and large lot quality conformance inspection |
|---|---|---|---|
| | Method | Conditions | |
| Subgroup 1 | | | 45 devices<br>c = 0 |
| Temperature cycling | 1051 | Test condition G, 500 cycles | |
| Hermetic seal<br>Fine leak<br>Gross leak | 1071 | <br>Test conditions G or H<br>Test conditions C or D | |
| Electrical measurements | | See table I, subgroup 2. | |
| Subgroup 2 1/ | | | 45 devices<br>c = 0 |
| Steady-state reverse bias | 1042 | Condition A, 1,000 hours. | |
| Electrical measurements | | See table I, subgroup 2. | |
| Steady-state gate bias | 1042 | Condition B, 1,000 hours. | |
| Electrical measurements | | See table I, subgroup 2. | |
| Subgroup 4 | | | Sample size<br>N/A |
| * Thermal impedance curves | | See MIL-PRF-19500. | |
| Subgroup 10 | | | 22 devices<br>c = 0 |
| Commutating diode for safe operating area test procedure for measuring dv/dt during reverse recovery of power MOSFET transistors or insulated gate bipolar transistors | 3476 | Test conditions shall be derived by the manufacturer | |

1/ A separate sample for each test shall be pulled.



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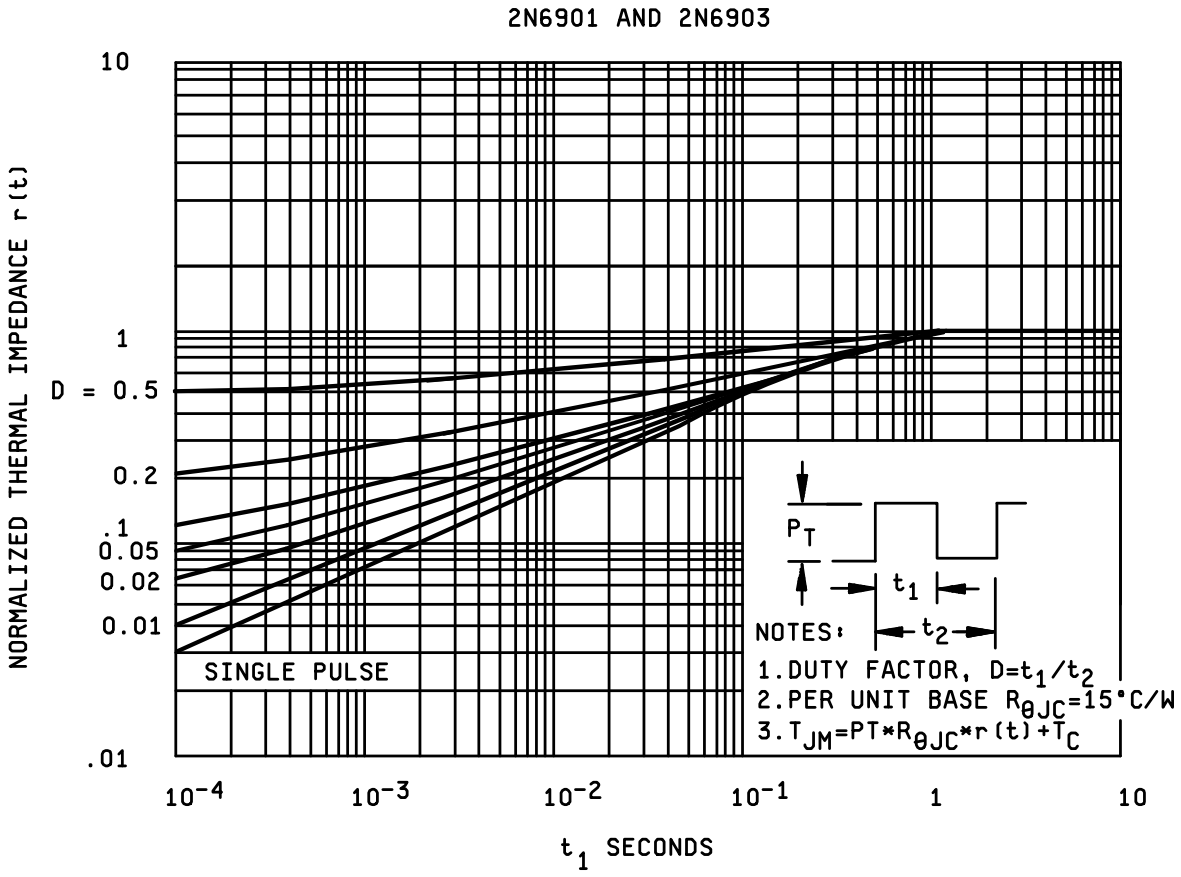



FIGURE 2. Transient thermal response.



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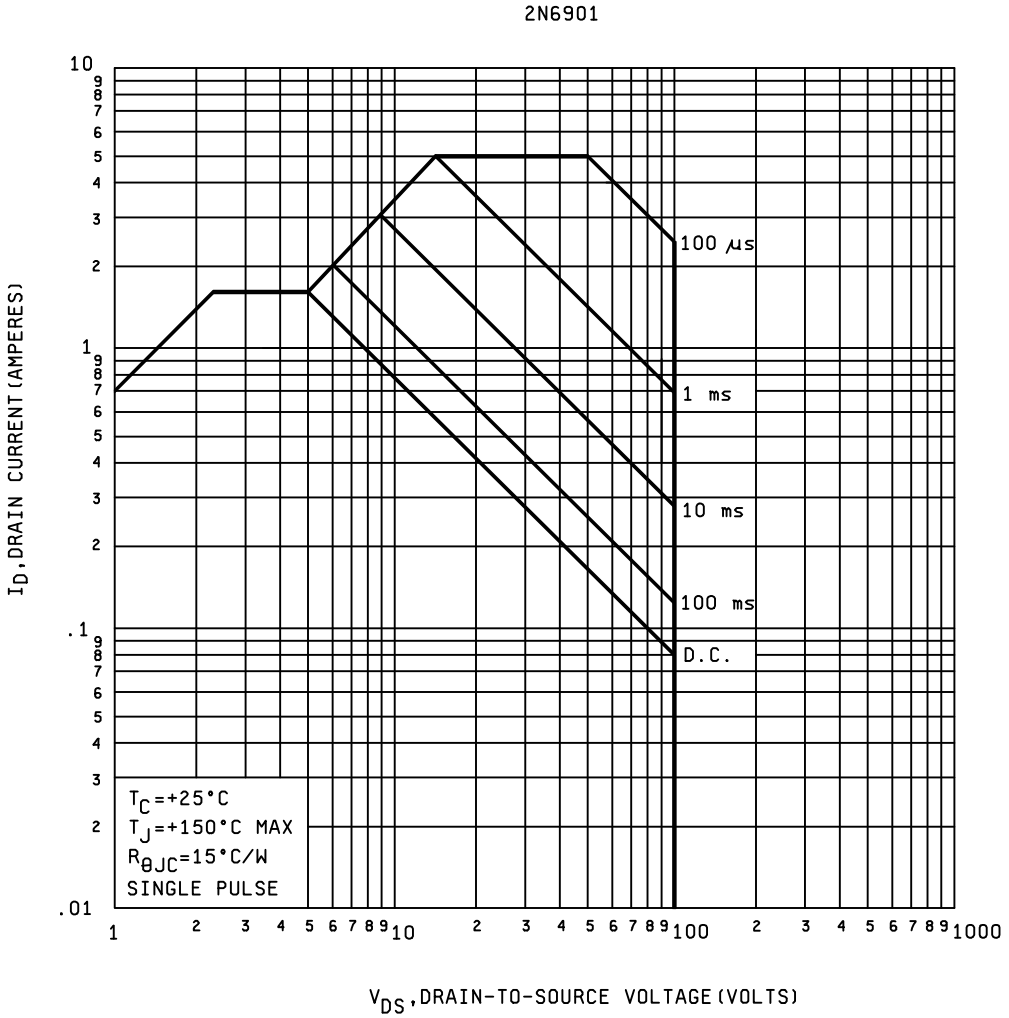



FIGURE 3. Maximum safe operating area.



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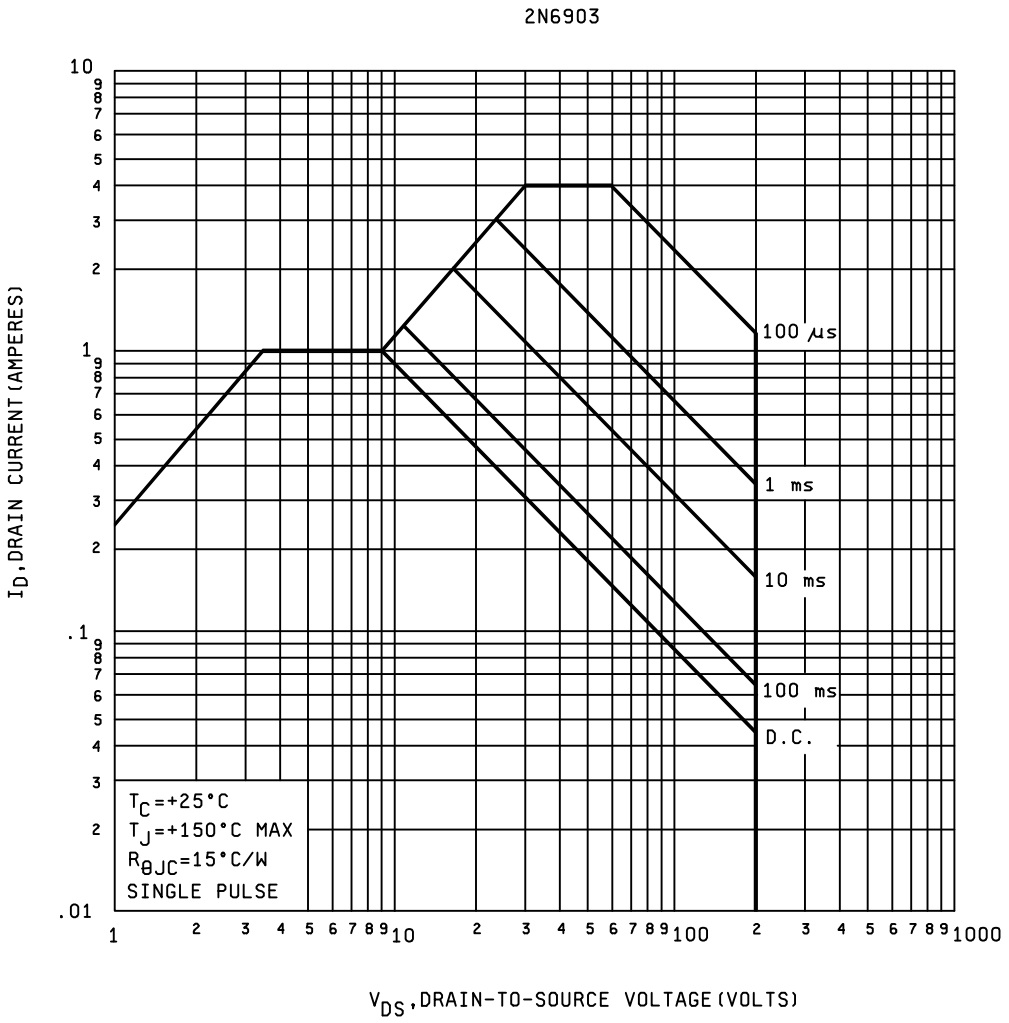



FIGURE 3. Maximum safe operating area - Continued.



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5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When packaging of materiel is to be performed by DoD or in-house contractor personnel, these personnel need to contact the responsible packaging activity to ascertain packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activities within the Military Service or Defense Agency, or within the Military Service's system commands. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6. NOTES

* (This section contains information of a general or explanatory nature that may be helpful, but is not mandatory. The notes specified in MIL-PRF-19500 are applicable to this specification.)

* 6.1 Intended use. Semiconductors conforming to this specification are intended for original equipment design applications and logistic support of existing equipment.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.

b. Packaging requirements (see 5.1).

c. Lead finish (see 3.4.1).

d. Product assurance level and type designator.

* 6.3 Qualification. With respect to products requiring qualification, awards will be made only for products which are, at the time of award of contract, qualified for inclusion in Qualified Manufacturers List (QML 19500) whether or not such products have actually been so listed by that date. The attention of the contractors is called to these requirements, and manufacturers are urged to arrange to have the products that they propose to offer to the Federal Government tested for qualification in order that they may be eligible to be awarded contracts or orders for the products covered by this specification. Information pertaining to qualification of products may be obtained from Defense Supply Center, Columbus, ATTN: DSCC/VQE, P.O. Box 3990, Columbus, OH 43218-3990 or e-mail vqe.chief@dla.mil. An online listing of products qualified to this specification may be found in the Qualified Products Database (QPD) at http://assist.daps.dla.mil .

6.4 Changes from previous issue. The margins of this specification are marked with asterisks to indicate where changes from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

Custodians:
Army - CR
Navy - EC
Air Force - 85
DLA - CC

Preparing activity:
DLA - CC

(Project 5961-2008-048)

Review activities:
Army - AR, MI, SM
Navy - AS, MC, OS, SH
Air Force - 19

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at http://assist.daps.dla.mil.



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